LAB – 1

1) Create Table employee with attributes firstname, lastname, SSN , address, salary, birthday, sex, supervisor SSN, department No.

```
CREATE TABLE Employee (
    Fname VARCHAR(20),
    Lname VARCHAR(20),
    SSN CHAR(9) PRIMARY KEY,
    Address VARCHAR(100),
    Salary NUMBER(10, 2),
    DOB DATE,
    Sex CHAR(1),
    SuperSSN CHAR(9),
    D.No NUMBER
);
```

OUTPUT:

Table created

2) Create a table department with attributes DNo, DName, Manager SSN, Mgr StartDate

```
CREATE TABLE Department (
    DNo NUMBER PRIMARY KEY,
    DName VARCHAR (30),
```

```
    MgrSSN CHAR (9),
    MgrStartDate DATE
);
```

OUTPUT:

Table created

3) Insert value in table

```
 INSERT INTO EMPLOYEE
    VALUES(
    'John',
    'Smith',
    '54545454',
    'Chandni Chowk Delhi',
    57555,
    TO_DATE('01-01-2001','DD-MM-YYYY'),
    'M',
   NULL,
   1
  );
```

OUTPUT:

1 row created.

4) Display all employee Information

```
SELECT * FROM EMPLOYEE;
```

OUTPUT:

```
FIRSTNAME
--------------------------------------------------
LASTNAME                                 SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
   SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
John
Smith                                    54545454
Chandni Chowk Delhi
    57555 01-JAN-01 M                    1
```

5) Display employee name along with SSN, SupervisorSSN

```
SQL> SELECT Firstname,lastname, SSN, SupervisorSSN
  2  from employee;
```

OUTPUT:

```
FIRSTNAME
--------------------------------------------------
LASTNAME                                 SSN       SUPERVISO
-------------------------------------------------- --------- ---------
John
```

Smith 54545454

6) Display the employee names where DOB is '29 Mar 1959'

```
SQL> SELECT Firstname,lastname
  from employee
  where birthday = TO_DATE('29-05-1959','dd-mm-yyyy');
```

OUTPUT:

no rows selected

ScreenShot

Command Prompt - SQLPLUS MIS247068/MIS247068@172.18.8.101/ORCLPDB1

```
SQL> INSERT INTO EMPLOYEE
  2  VALUES(
  3  'John',
  4  'Smith',
  5  '54545454',
  6  'Chandni Chowk Delhi',
  7  57555,
  8  TO_DATE('01-01-2001','DD-MM-YYYY'),
  9  'M',
 10  NULL,
 11  1
 12  );

1 row created.

SQL> SELECT * FROM EMPLOYEE
  2  ;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
--------- --------- - --------- ------------
John
Smith                                              54545454
Chandni Chowk Delhi
     57555 01-JAN-01 M                     1


SQL> SELECT Firstname,lastname, SSN, Supervisor
  2  from employee;
SELECT Firstname,lastname, SSN, Supervisor
                                *
ERROR at line 1:
ORA-00904: "SUPERVISOR": invalid identifier


SQL> SELECT Firstname,lastname, SSN, SupervisorSSN
  2  from employee;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN       SUPERVISO
-------------------------------------------------- --------- ---------
John
Smith                                              54545454
```

LAB – 2

1) Display salary of the employee without duplication

```
SQL> SELECT DISTINCT SALARY
   FROM EMPLOYEE;
```

OUTPUT:

```
   SALARY
----------
    65000
    57555
```

2) Display the MgrSSN , MgrStartDate of the manager of 'Finance' Department

```
SQL> SELECT MANAGERSSN, MGRSTARTDate
   FROM DEPARTMENT
   WHERE DNAME = 'Finance';
```

OUTPUT:

```
MANAGERSS MGRSTARTD
--------- ---------
323232323 11-NOV-22
```

3) Modify the department no of employee having Fname as 'Joyce' to 5

```
update employee
   set DepartmentNo = 5
   where Firstname = 'Mike';
```

OUTPUT:

1 row updated.

4) Alter table department add column deptPhoneNo

```
SQL> ALter table Department
   ADD (DepartmentPhoneNo NUMBER);
```

OUTPUT:

Table altered.

5) Alter table department modify column deptPhoneNo size

```
 SQL> ALter table Department
  2  MODIFY (DepartmentPhoneNo NUMBER(15));
```

OUTPUT:

Table altered.

6)Alter table department rename column deptPhoneNo to PhNo

SQL> ALter table Department

RENAME COLUMN DepartmentPhoneNo TO PHNO;

OUTPUT:

Table altered.

7) Rename table department to Dept

SQL> RENAME DEPARTMENT TO DEPT;

OUTPUT:

Table renamed.

8) Alter employee remove column PhNO

SQL> ALter table DepT

DROP COLUMN PHNO;

OUTPUT:

Table altered.

ScreenShot

```
SQL> SELECT DISTINCT SALARY
  2  FROM EMPLOYEE;

    SALARY
----------
     65000
     57555
```

```
SQL> SELECT MANAGERSSN, MGRSTARTDate
  2  FROM DEPARTMENT
  3  WHERE DNAME = 'Finance';

MANAGERSS MGRSTARTD
--------- ---------
323232323 11-NOV-22

SQL> update employee
  2  set DNo = 5
  3  where Firstname = 'Mike';
set DNo = 5
    *
ERROR at line 2:
ORA-00904: "DNO": invalid identifier


SQL> update employee
  2  where Firstname = 'Mike';
where Firstname = 'Mike'
*
ERROR at line 2:
ORA-00971: missing SET keyword


SQL> update employee
  2  set DNo = 5
  3  ';
ERROR:
ORA-01756: quoted string not properly terminated


SQL> update employee
  2  set DepartmentNo = 5
  3  where Firstname = 'Mike';

1 row updated.
```

```
SQL> ALter table Department
  2  ADD (DepartmentPhoneNo NUMBER);

Table altered.

SQL> ALter table Department
  2  MODIFY (DepartmentPhoneNo NUMBER);

Table altered.

SQL> ALter table Department
  2  MODIFY (DepartmentPhoneNo NUMBER(15));

Table altered.

SQL> ALter table Department
  2  RENAME DepartmentPhoneNo TO PHNO;
RENAME DepartmentPhoneNo TO PHNO
       *
ERROR at line 2:
ORA-14155: missing PARTITION, PARTITIONS, SUBPARTITION, or SUBPARTITIONS
keyword


SQL> ALter table Department
  2  RENAME COLUMN DepartmentPhoneNo TO PHNO;

Table altered.

SQL> RENAME DEPARTMENT TO DEPT;

Table renamed.

SQL> ALter table DepT
  2  DROP COLUMN PHNO
  3  ;

Table altered.
```

LAB – 3

1) Create a table COPYOFDEPT as a copy of the table DEPT.

SQL> CREATE TABLE COPYOFDEPT

AS SELECT * FROM DEPT;

OUTPUT:

Table created.

2) Delete all the rows from COPYOFDEPT table.

SQL> DELETE FROM COPYOFDEPT;

OUTPUT:

2 rows deleted.

3) Remove COPYOFDEPT table.

SQL> DROP TABLE COPYOFDEPT;

OUTPUT:

Table dropped

4) Add Foreign Keys using Alter Table.

SQL> ALTER TABLE EMPLOYEE

ADD CONSTRAINT fk_dept

FOREIGN KEY (DEPARTMENTNO) REFERENCES dept(dNo);

OUTPUT:

Table altered.

5) Drop Foreign key defined on SuperSSN and add it using Alter table command.

SQL> ALTER TABLE EMPLOYEE

Drop CONSTRAINT fk_superssn;

OUTPUT:

Table altered.

6) Find the employee names having salary greater than Rs.25000.

SQL> SELECt FIRstname from EMPLOYEE WHERE Salary > 25000;

OUTPUT:

FIRSTNAME

John

Jane

Mike

```
SQL> CREATE TABLE COPYOFDEPT
  2  AS SELECT * FROM DEPT;

Table created.

SQL> DELETE FROM COPYOFDEPT;

2 rows deleted.

SQL> DROP TABLE COPYDEPT;
DROP TABLE COPYDEPT
           *
ERROR at line 1:
ORA-00942: table or view does not exist


SQL> DROP TABLE COPYOFDEPT;

Table dropped.
```

```
SQL> ALTER TABLE EMPLOYEE
  2  ADD CONSTRAINT fk_dept
  3  FOREIGN KEY (DEPARTMENTNO) REFERENCES dept(dNo);

Table altered.

SQL> ALTER TABLE EMPLOYEE
  2  ADD CONSTRAINT fk_superssn
  3  FOREIGN KEY (supervisorSSn) REFERENCES dept(managerssn);
FOREIGN KEY (supervisorSSn) REFERENCES dept(managerssn)
                                           *
ERROR at line 3:
ORA-02270: no matching unique or primary key for this column-


SQL> ALTER TABLE EMPLOYEE
  2  Drop CONSTRAINT fk_superssn;
Drop CONSTRAINT fk_superssn
                *
ERROR at line 2:
ORA-02443: Cannot drop constraint  - nonexistent constraint


SQL> SELECt FIRstname from EMPLOYEE WHERE Salary > 25000;

FIRSTNAME
--------------------------------------------------
John
Jane
Mike
```

LAB – 4

1) Find the employee names whose salary lies in the range between 30000 and 70000

SQL> SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE

2 WHERE SALARY BETWEEN 30000 AND 70000;

OUTPUT:

FIRSTNAME

LASTNAME

John

Smith

Jane

Doe

Mike

Johnson

```
SQL> SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE
  2  WHERE SALARY BETWEEN 30000 AND 70000;

FIRSTNAME
--------------------------------------------------
LASTNAME
--------------------------------------------------
John
Smith

Jane
Doe

Mike
Johnson
```

2) Find the employees who have no supervisor.

```
SQL> SELECT * FROM EMPLOYEE
        WHERE SUPERVISORSSN IS NULL;
```

OUTPUT:

```
FIRSTNAME
--------------------------------------------------
LASTNAME                                 SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
John
Smith                                    54545454
Chandni Chowk Delhi
    57555 01-JAN-01 M                    1


FIRSTNAME
--------------------------------------------------
LASTNAME                                 SSN
-------------------------------------------------- ---------
ADDRESS
```

SALARY BIRTHDAY S SUPERVISO DEPARTMENTNO

--- --- - --- ---

Jane

Doe 123456789

456 MG Road Mumbai

65000 15-MAY-95 F 2

```
SQL> SELECT * FROM EMPLOYEE
  2  WHERE SUPERVISORSSN IS NULL;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
John
Smith                                              54545454
Chandni Chowk Delhi
     57555 01-JAN-01 M                     1


FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
Jane
Doe                                                123456789
456 MG Road Mumbai
     65000 15-MAY-95 F                     2
```

3)Display the bdate of all employees in the format ‘DDthMonthYYYY’.

```
SELECT TO_CHAR(BIRTHDAY, 'DDth Month YYYY') AS BDAY
  FROM EMPLOYEE;
```

OUTPUT:

```
BDAY
---------------------------------------------
01ST January   2001
15TH May       1995
22ND August    1988
```

```
SQL> SELECT FIRSTNAME,LASTNAME, TO_CHAR(BIRTHDAY, 'DDth Month YYYY') AS BDAY
  2  FROM EMPLOYEE;

FIRSTNAME
--------------------------------------------------
LASTNAME
--------------------------------------------------
BDAY
----------------------------------------------
John
Smith
01ST January   2001

Jane
Doe
15TH May       1995

FIRSTNAME
--------------------------------------------------
LASTNAME
--------------------------------------------------
BDAY
----------------------------------------------

Mike
Johnson
22ND August    1988


SQL> SELECT TO_CHAR(BIRTHDAY, 'DDth Month YYYY') AS BDAY
  2  FROM EMPLOYEE;

BDAY
----------------------------------------------
01ST January   2001
15TH May       1995
22ND August    1988
```

4) Display the employee names whose bdate is on or before 1978.

SELECT FIRSTNAME, LASTNAME, FROM EMPLOYEE

WHERE BIRTHDAY <= TO_DATE(‘31-DEC-1978’, ‘DD-MM-YYYY’);

OUTPUT:

no rows selected

5) Display the employee names having ‘salt lake’ in their address.

SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE

WHERE LOWER(ADDRESS) LIKE '%salt lake%';

OUTPUT:

no rows selected

```
SQL> SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE
  2  WHERE BIRTHDAY <= TO_DATE('31-DEC-1978', 'DD-MM-YYYY');

no rows selected

SQL> SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE
  2  WHERE LOWER(ADDRESS) LIKE '%salt lake%';

no rows selected
```

6) Display the department name that starts with ‘M’.

SELECT DNAME from DEPT

WHERE DNAME LIKE 'M%';

OUTPUT:

DNAME

Marketing

7) Display the department names that ends with ‘E’.

SELECT DNAME from DEPT

WHERE DNAME LIKE '%E';

OUTPUT:

no rows selected

8) Display the names of all the employees having supervisor with any of the following SSN 554433221, 333445555.

SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE

WHERE SUPERVISORSSN IN(554433221, 333445555);

OUTPUT:

no rows selected

9) Display all the department names in upper case and lower case.

SELECT UPPER(DNAME) AS UPPERCASED, LOWER(DNAME) AS LOWERCASED

FROM DEPT;

OUTPUT:

UPPERCASED

LOWERCASED

FINANCE

finance

MARKETING

marketing

ADMIN

admin

10) Display the first four characters and last four of the department names using substring function.

SQL> SELECT SUBSTR(DNAME , 1,4) AS FIRST_FOUR, SUBSTR(DNAME, LENGTH(DNAME) -3 , 4) AS LAST_FOUR

FROM DEPT;

OUTPUT:

FIRST_FOUR LAST_FOUR

--- ---

Fina ance

Mark ting

Admi dmin

Command Prompt - SQLPLUS MIS247068/MIS247068@172.18.8.101/ORCLPDB1

```
SQL> SELECT DNAME from DEPT
  2  WHERE DNAME LIKE 'M%';

DNAME
--------------------------------------------------
Marketing

SQL> DESC DEPT;
SQL> SELECT DNAME from DEPT
  2  WHERE DNAME LIKE '%E';

no rows selected

SQL> SELECT FIRSTNAME, LASTNAME FROM EMPLOYEE
  2  WHERE SUPERVISORSSN IN(554433221, 333445555);

no rows selected

SQL> SELECT UPPER(DNAME) AS UPPERCASED, LOWER(DNAME) AS LOWERCASED
  2  FROM DEPT;

UPPERCASED
--------------------------------------------------
LOWERCASED
--------------------------------------------------
FINANCE
finance

MARKETING
marketing

ADMIN
admin


SQL> SELECT SUBSTR(DNAME , 1,4) AS FIRST_FOUR, SUBSTR(DNAME, LENGTH(DNAME) -3 , 4) AS LAST_FOUR,
  2  FROM DEPT;
FROM DEPT
*
ERROR at line 2:
ORA-00936: missing expression


SQL> SELECT SUBSTR(DNAME , 1,4) AS FIRST_FOUR, SUBSTR(DNAME, LENGTH(DNAME) -3 , 4) AS LAST_FOUR
  2  FROM DEPT;

FIRST_FOUR       LAST_FOUR
---------------- ----------
Fina             ance
Mark             ting
Admi             dmin
```

LAB 5

1) Display the substring of the Address (starting from 5th position to 11 th position) of all employees.

```
SELECT SUBSTR(ADDRESS , 5,7)
  FROM EMPLOYEE;
```

OUTPUT:

```
SUBSTR(ADDRESS,5,7)
----------------------------
dni Cho
MG Road
Park St
```

2) Display the Mgrstartdate on adding three months to it.

```
SQL> SELECT ADD_MONTHS(MGRSTARTDATE,
FROM DEPT;
```

OUTPUT:

```
ADD_MONTH
---------
11-FEB-23
```

21-JUL-21

```
SQL> SELECT SUBSTR(DNAME , 1,4) AS FIRST_FOUR, SUBSTR(DNAME, LENGTH(DNAME) -3 , 4) AS LAST_FOU
  2  FROM DEPT;

FIRST_FOUR       LAST_FOUR
---------------- ----------------
Fina             ance
Mark             ting
Admi             dmin

SQL> SELECT SUBSTR(ADDRESS , 5,7)
  2  FROM EMPLOYEE;

SUBSTR(ADDRESS,5,7)
---------------------------
dni Cho
MG Road
Park St

SQL> SELECT ADD_MONTHS(MGRSTARTDATE, 3)
  2  FROM DEPT;

ADD_MONTH
---------
11-FEB-23

21-JUL-21
```

3) Display the age of all the employees rounded to two digits.

SELECT ROUND((SYSDATE - BIRTHDAY)/ 365,2) AS AGE

FROM EMPLOYEE;

OUTPUT:

AGE

25.19

30.83

37.56

4) Find the last day and next day of the month in which each manager has joined.

SELECT LAST_DAY(MGRSTARTDATE) AS LASTDAY,
LAST_DAY(MGRSTARTDATE)+ 1 AS NEXTDAY

FROM DEPT;

OUTPUT:

LASTDAY NEXTDAY

--------- ---------

30-NOV-22 01-DEC-22

30-APR-21 01-MAY-21

```
SQL> SELECT ROUND((SYSDATE - BIRTHDAY)/ 365,2 ) AS AGE
  2  FROM EMPLOYEE;

       AGE
----------
     25.19
     30.83
     37.56

SQL> SELECT LAST_DAY(MGRSTARTDATE) AS LASTDAY, LAST_DATE(MGRSTARTDATE)+ 1 AS NEXTDA
  2  FROM DEPT;
SELECT LAST_DAY(MGRSTARTDATE) AS LASTDAY, LAST_DATE(MGRSTARTDATE)+ 1 AS NEXTDAY
                                          *
ERROR at line 1:
ORA-00904: "LAST_DATE": invalid identifier


SQL> SELECT LAST_DAY(MGRSTARTDATE) AS LASTDAY, LAST_DAY(MGRSTARTDATE)+ 1 AS NEXTDAY
  2  FROM DEPT;

LASTDAY   NEXTDAY
--------- ---------
30-NOV-22 01-DEC-22

30-APR-21 01-MAY-21
```

5) Print a substring from the string ‘Harini’.

SELECT SUBSTR('Harini', 2,4)
FROM DUAL;

OUTPUT:

SUBS

---

arin

6) Replace the string ‘ni’ from ‘Harini’ by ‘sh’.

```
SELECT REPLACE('Harini','ni','sh')
  2  FROM DUAL;
```

OUTPUT:

REPLACE

Harish

7)Print the length of all the department names.

```
 SELECT DNAME, LENGTH(DNAME) FROM DEPT;
```

OUTPUT:

DNAME	LENGTH(DNAME)
Finance	7
Marketing	9
Admin	5

8) Print the system date in the format 25 th May 2007.

```
SELECT TO_CHAR(SYSDATE, 'DDTH MONTH YYYY')
FROM DUAL;
```

OUTPUT:

```
TO_CHAR(SYSDATE,'DDTHMONTHYYYY')
--------------------------------------------
05TH MARCH     2026
```

9) Display the date after 10 months from current date.

```
SELECT ADD_MONTHS(SYSDATE,10)
FROM DUAL;
```

OUTPUT:

```
ADD_MONTH
---------
05-JAN-27
```

10) Display the next occurrence of Friday in this month.

```
SELECT NEXT_DAY(SYSDATE, 'FRIDAY')
  2  FROM DUAL;
```

OUTPUT:

```
NEXT_DAY(
```

06-MAR-26

```
SQL> SELECT SUBSTR('Harini', 2,4)
  2  FROM DUAL;

SUBS
----
arin

SQL> SELECT REPLACE('Harini','ni','sh')
  2  FROM DUAL;

REPLAC
------
Harish

SQL> SELECT DNAME, LENGTH(DNAME) FROM DEPT
  2  ;

DNAME                                                  LENGTH(DNAME)
------------------------------------------------------ -------------
Finance                                                            7
Marketing                                                          9
Admin                                                              5

SQL> SELECT TO_CHAR(SYSDATE, 'DDTH MONTH YYYY')
  2  FROM DUAL;

TO_CHAR(SYSDATE,'DDTHMONTHYYYY')
--------------------------------------------
05TH MARCH     2026

SQL> SQL> SELECT ADD_MONTHS(SYSDATE,10)
SP2-0734: unknown command beginning "SQL> SELEC..." - rest of lin
SQL> SELECT ADD_MONTHS(SYSDATE,10)
  2  FROM DUAL;

ADD_MONTH
---------
05-JAN-27

SQL> SELECT NEXT_DAY(SYSDATE, 'FRIDAY')
  2  FROM DUAL;

NEXT_DAY(
---------
06-MAR-26
```

LAB – 6

1) How many different departments are there in the 'employee' table

SELECT COUNT(DISTINCT DEPARTMENTNO)

FROM EMPLOYEE;

OUTPUT:

COUNT(DISTINCTDEPARTMENTNO)

3

```
SQL> SELECT COUNT(DISTINCT DEPARTMENTNO)
  2  FROM EMPLOYEE;

COUNT(DISTINCTDEPARTMENTNO)
---------------------------
                          3
```

2) For each department display the minimum and maximum employee salaries

SELECT DEPARTMENTNO, MIN(SALARY) , MAX(SALARY)
FROM EMPLOYEE GROUP BY DEPARTMENTNO;

OUTPUT:

DEPARTMENTNO MIN(SALARY) MAX(SALARY)

---------- ---------- ----------

1	57555	57555
2	65000	65000
5	57555	57555

```
SQL> SELECT DEPARTMENTNO, MIN(SALARY) , MAX(SALARY) FROM EMPLOYEE GROUP BY DEPARTMENTN

DEPARTMENTNO MIN(SALARY) MAX(SALARY)
------------ ----------- -----------
           1       57555       57555
           2       65000       65000
           5       57555       57555
```

3) Print the average annual salary.

SELECT AVG(SALARY) * 12 FROM EMPLOYEE;

OUTPUT:

AVG(SALARY)*12

720440

OR

SELECT AVG(SALARY) * 12 AS "Annual_Sal" FROM EMPLOYEE;

OUTPUT:

Annual_Sal

720440

```
SQL> SELECT AVG(SALARY) * 12 FROM EMPLOYEE;

AVG(SALARY)*12
--------------
        720440

SQL> SELECT AVG(SALARY) * 12 AS "Annual_Sal" FROM EMPLOYEE;

Annual_Sal
----------
    720440
```

4) Count the number of employees over 30 age.

SELECT COUNT(*) FROM EMPLOYEE WHERE (SYSDATE - BIRTHDAY)/365 > 30;

OUTPUT:

```
  COUNT(*)
---------
       2
```

```
SQL> SELECT COUNT(*) FROM EMPLOYEE WHERE (SYSDATE - BIRTHDAY)/365  > 30

  COUNT(*)
---------
         2
```

5) Print the Department name and average salary of each department.

SELECT D.DNAME,

AVG(E.SALARY)

FROM EMPLOYEE E

JOIN DEPT D

ON E.DEPARTMENTNO = D.DNO

GROUP BY D.DNAME;

```
SQL> SELECT D.DNAME,
  2  AVG(E.SALARY)
  3  FROM EMPLOYEE E
  4  JOIN DEPT D
  5  ON E.DEPARTMENTNO = D.DNO
  6  GROUP BY D.DNAME;

DNAME                                              AVG(E.SALARY)
-------------------------------------------------- -------------
Admin                                                      57555
Marketing                                                  57555
Finance                                                    65000
```

6) Create a view to display the employee details who is working in IT department.

CREATE VIEW IT_EMPLOYEES AS

SELECT *

FROM EMPLOYEE

WHERE DEPARTMENTNO =

(

SELECT DNO

FROM DEPT

WHERE DNAME='IT'

);

```
SQL> CREATE VIEW IT_EMPLOYEES AS
  2  SELECT *
  3  FROM EMPLOYEE
  4  WHERE DEPARTMENTNO =
  5  (
  6  SELECT DNO
  7  FROM DEPT
  8  WHERE DNAME='IT'
  9  );

View created.
SQL> Select * from IT_EMPLOYEES;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
--------- --------- - --------- ------------
Alice
Williams                                           110099221
101 Tech Park Bangalore
     54212 12-MAR-88 F                     6
```

7) Create a logical table to store employee details who is getting salary more than 10000.

```
CREATE VIEW HIGH_SALARY_EMP AS
  2  SELECT *
  3  FROM EMPLOYEE
```

```
4  WHERE SALARY > 10000;
```

```
SQL> CREATE VIEW HIGH_SALARY_EMP AS
  2  SELECT *
  3  FROM EMPLOYEE
  4  WHERE SALARY > 10000;

View created.

SQL> SELECT * FROM HIGH_salary_EMP;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
John
Smith                                              54545454
Chandni Chowk Delhi
     57555 01-JAN-01 M                      1


FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
Jane
Doe                                                123456789
456 MG Road Mumbai
     65000 15-MAY-95 F                      2


FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
Mike
Johnson                                            987654321
789 Park Street Kolkata
     57555 22-AUG-88 M 54545454             5


FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
```

8) Create a table to store the employees details based on the department no

```
CREATE TABLE EMP_DEPT5 AS
 SELECT *
FROM EMPLOYEE
 WHERE DEPARTMENTNO = 5;
```

OUTPUT:

Table created.

```
SQL> CREATE TABLE EMP_DEPT5 AS
  2  SELECT *
  3  FROM EMPLOYEE
  4  WHERE DEPARTMENTNO = 5;
```

```
SQL> SELECT * FROM EMP_DEPT5;

FIRSTNAME
--------------------------------------------------
LASTNAME                                           SSN
-------------------------------------------------- ---------
ADDRESS
--------------------------------------------------------------------------------
    SALARY BIRTHDAY  S SUPERVISO DEPARTMENTNO
---------- --------- - --------- ------------
Mike
Johnson                                            987654321
789 Park Street Kolkata
     57555 22-AUG-88 M 54545454            5
```

9)List the names of all managers who have no dependents.

SELECT FIRSTNAME, LASTNAME

FROM EMPLOYEE

WHERE SSN IN (SELECT MANAGERSSN FROM DEPT)

AND SSN NOT IN (SELECT ESSN FROM DEPENDENT);

```
SQL> create table dependent(
  2  ESSN CHAR(9),
  3  DEPENDENT_NAME VARCHAR2(50),
  4  SEX CHAR(1),
  5  DOB DATE,
  6  RELATIONSHIP VARCHAR2(20),
  7  PRIMARY KEY (ESSN, DEPENDENT_NAME),
  8  FOREIGN KEY (ESSN) REFERENCES EMPLOYEE(SSN));

Table created.
```

```
SQL> INSERT INTO dependent (ESSN, DEPENDENT_NAME, SEX, DOB, RELATIONSHIP)
  2  VALUES ('123456789', 'Ryan', 'M', TO_DATE('10-01-2016', 'DD-MM-YYYY'), 'SON');

1 row created.

SQL> SELECT * FROM depeNDENT;

ESSN      DEPENDENT_NAME                                       S DOB
--------- ---------------------------------------------------- - ---------
RELATIONSHIP
--------------------
123456789 Ryan                                                 M 10-JAN-16
SON
```

```
SQL> SELECT FIRSTNAME, LASTNAME
  2  FROM EMPLOYEE
  3  WHERE SSN IN (SELECT MANAGERSSN FROM DEPT)
  4  AND SSN NOT IN (SELECT ESSN FROM DEPENDENT);

no rows selected
```

10) List the employee's names and the department names if they happen to manage a department.

SELECT E.FIRSTNAME,

E.LASTNAME,

D.DNAME

FROM EMPLOYEE E

JOIN DEPT D

ON E.SSN = D.MANAGERSSN;

```
SQL> SELECT E.FIRSTNAME,
  2  E.LASTNAME,
  3  D.DNAME
  4  FROM EMPLOYEE E
  5  JOIN DEPT D
  6  ON E.SSN = D.MANAGERSSN;

no rows selected
```

LAB 7

1) Write a PL/SQL program to display "Hello World".

CODE:

```
DECLARE
BEGIN
DBMS_OUTPUT.PUT_LINE('Hello World');
END;
/
```

OUTPUT:

queries.sql +

Week 7

```
DECLARE
BEGIN
DBMS_OUTPUT.PUT_LINE('Hello World');
END;
/
```

STDIN

Input for the program (Optional

Output:

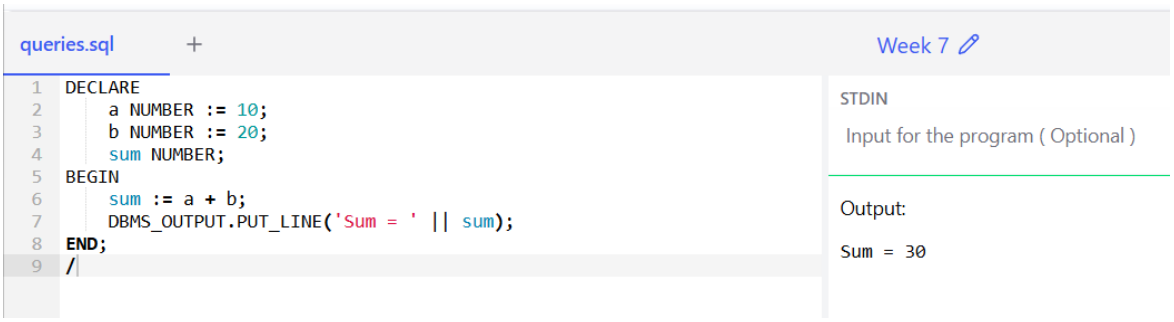
```
Hello World
```

2)Write a PL/SQL program to add two numbers and display the result.

CODE:

```
 DECLARE
    a NUMBER := 10;
    b NUMBER := 20;
    sum NUMBER;
BEGIN
    sum := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum);
END;
/
```

OUTPUT:



3)Write a PL/SQL program to check whether a number is prime.

CODE:

```
DECLARE
    n NUMBER := 7;
    i NUMBER;
    flag BOOLEAN := TRUE;
```

```
BEGIN
    IF n <= 1 THEN
        flag := FALSE;
    ELSE
        FOR i IN 2..n-1 LOOP
            IF MOD(n, i) = 0 THEN
                flag := FALSE;
                EXIT;
            END IF;
        END LOOP;
    END IF;

    IF flag THEN
        DBMS_OUTPUT.PUT_LINE(n || ' is Prime');
    ELSE
        DBMS_OUTPUT.PUT_LINE(n || ' is Not Prime');
    END IF;
END;
/
```

OUTPUT:

queries.sql

Week 7

```sql
DECLARE
    n NUMBER := 7;
    i NUMBER;
    flag BOOLEAN := TRUE;
BEGIN
    IF n <= 1 THEN
        flag := FALSE;
    ELSE
        FOR i IN 2..n-1 LOOP
            IF MOD(n, i) = 0 THEN
                flag := FALSE;
                EXIT;
            END IF;
        END LOOP;
    END IF;

    IF flag THEN
        DBMS_OUTPUT.PUT_LINE(n || ' is Prime');
    ELSE
        DBMS_OUTPUT.PUT_LINE(n || ' is Not Prime');
    END IF;
END;
/
```

STDIN

Input for the program (Optional)

Output:

```
7 is Prime
```

4)Write a PL/SQL program to find the factorial of a number.

CODE:

```sql
DECLARE
    n NUMBER := 5;
    fact NUMBER := 1;
BEGIN
    FOR i IN 1..n LOOP
        fact := fact * i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Factorial = ' || fact);
END;
```

/

OUTPUT:

queries.sql +

Week 7

```
DECLARE
    n NUMBER := 5;
    fact NUMBER := 1;
BEGIN
    FOR i IN 1..n LOOP
        fact := fact * i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Factorial = ' || fact);
END;
/
```

STDIN

Input for the program (Optional)

Output:

```
Factorial = 120
```

5) Write a PL/SQL program to print the multiplication table of a number.

CODE:

```
DECLARE
   n NUMBER := 5;
BEGIN
   FOR i IN 1..10 LOOP
      DBMS_OUTPUT.PUT_LINE(n || ' x ' || i || ' = ' || (n*i));
   END LOOP;
END;
```

/

OUTPUT:

queries.sql +

Week 7

```
DECLARE
    n NUMBER := 5;
BEGIN
    FOR i IN 1..10 LOOP
        DBMS_OUTPUT.PUT_LINE(n || ' x ' || i || ' = ' || (n*i));
    END LOOP;
END;
/
```

STDIN

Input for the program (Optional)

Output:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

6) Write a PL/SQL program to reverse a number.

CODE:

```
DECLARE
    n NUMBER := 1234;
    rev NUMBER := 0;
    rem NUMBER;
BEGIN
    WHILE n > 0 LOOP
        rem := MOD(n, 10);
        rev := rev * 10 + rem;
        n := TRUNC(n / 10);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Reversed Number = ' || rev);
```

END;

/

OUTPUT:

queries.sql +

Week 7

```
DECLARE
    n NUMBER := 1234;
    rev NUMBER := 0;
    rem NUMBER;
BEGIN
    WHILE n > 0 LOOP
        rem := MOD(n, 10);
        rev := rev * 10 + rem;
        n := TRUNC(n / 10);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Reversed Number = ' || rev);
END;
/
```

STDIN

Input for the program (Optional)

Output:

```
Reversed Number = 4321
```

7) Write a PL/SQL program to generate Fibonacci series.

CODE:

```
DECLARE
    n NUMBER := 10;
    a NUMBER := 0;
    b NUMBER := 1;
    c NUMBER;
BEGIN
    DBMS_OUTPUT.PUT_LINE(a);
    DBMS_OUTPUT.PUT_LINE(b);
```

```
    FOR i IN 3..n LOOP
        c := a + b;
        DBMS_OUTPUT.PUT_LINE(c);
        a := b;
        b := c;
    END LOOP;
END;
/
```

OUTPUT:

queries.sql + Week 7

```sql
DECLARE
    n NUMBER := 10;
    a NUMBER := 0;
    b NUMBER := 1;
    c NUMBER;
BEGIN
    DBMS_OUTPUT.PUT_LINE(a);
    DBMS_OUTPUT.PUT_LINE(b);

    FOR i IN 3..n LOOP
        c := a + b;
        DBMS_OUTPUT.PUT_LINE(c);
        a := b;
        b := c;
    END LOOP;
END;
/
```

STDIN

Input for the program (Optional)

Output:

```
0
1
1
2
3
5
8
13
21
34
```